EE605 - Error Correcting Codes	September 4th, 2023

Lecture 11

Lecturer: Nikhil Karamchandani *Scribe(s): (Shashwat Gupta, Kunal Randad)*

1 Gilbert-Varshamov bound

Theorem 1 $\exists$ *(n,k) code such that*

$$2^k \geq \frac{2^n}{Vol_2(n,d-1)} \tag{1}$$

$$R = \frac{k}{n} \geq \frac{n - \log_2(\frac{2^n}{Vol_2(n,d-1)})}{n} \geq \lim_{n\to\infty} \frac{n - n \times H(\frac{d-1}{n})}{n} = 1 - H(\delta) \tag{2}$$

$$R \geq 1 - H(\delta) \tag{3}$$

Proof

$$\begin{aligned}\left(\frac{t}{n}\right)^n \left(1-\frac{t}{n}\right)^{n-t} \sum_{i=0}^{t} \binom{n}{i} &\leq \left(\frac{t}{n}\right)^t \left(1-\frac{t}{n}\right)^{n-t} \sum_{i=0}^{t} \binom{n}{i} \left(\frac{n-t}{t}\right)^{t-i} \forall\ t \leq \frac{n}{2} \\ &= \sum_{i=0}^{t} \binom{n}{i} \left(\frac{t}{n}\right)^i \left(1-\frac{t}{n}\right)^{n-i} \leq 1\end{aligned}$$

Using the above equations, we can write

$$\mathrm{Vol}_2(n,t) \times 2^{-nH(\frac{t}{n})} \leq 1 \quad \forall\ t \leq \frac{n}{2}$$

$$\mathrm{Vol}_2(n,t) \leq 2^{nH(\frac{t}{n})} \quad \forall\ t \leq \frac{n}{2} \tag{4}$$

From equation 1,

$$k \geq n - \log_2(\frac{2^n}{\mathrm{Vol}_2(n,d-1)}) \tag{5}$$

equation (2) and (3) follow from substituting equation (4) in (5) ■

2 Plotkin bound

Theorem 2 *Any (n, k, d) code over $GF(2)$ with $n > \frac{d}{2}$ satisfies the inequality:*

$$A_2(n, d) \leq \frac{2d}{2d - n}$$

Asymptotically, this is equivalent to:

$$R = \lim_{n\to\infty} \frac{\log A_2(n, d)}{n} \leq \lim_{n\to\infty} \frac{\log(\frac{2d}{2d-n})}{n} = 0 \qquad \textit{for } \delta > \frac{1}{2}$$

Proof Arrange the m code words along the rows of an $m \times n$ matrix.

$$\begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots \\ X_1 & X_2 & .... & X_{n-1} & X_n \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Denote by x_i the number of 1's in the i^{th} column.

$$\sum_{i \neq j} d(\mathbf{c_i}, \mathbf{c_j}) = \sum_{k=1}^{n} \binom{x_k}{1}\binom{m - x_k}{1} = \sum_{k=1}^{n} x_k(m - x_k)$$

The LHS is lower bounded by the product of the minimum possible distance between two codewords (d) and the number of codeword pairs. The RHS is upper bounded by the maximum possible value of the quadratic.

$$\binom{m}{2} d \leq \frac{nm^2}{4}$$
$$m \leq \frac{2d}{2d - n} \qquad \text{for } d > \frac{n}{2}$$

■

2.1 General Plotkin bound

This generalizes the statement of the Plotkin bound to codes with any value of normalized distance.

Theorem 3 *For a code with asymptotic rate R and asymptotic normalized distance d*

$$R \leq 1 - 2\delta$$

Proof Define a function $f : C \to \{0, 1\}^{n-2d+1}$ as:

$$f(c_1c_2 \dots c_n) = (c_1 \dots c_{n-2d+1})$$

Define set C_x for some $x \in \{0, 1\}^{n-2d+1}$ containing the suffixes of the codewords with prefix x:

$$C_x = \{(c_{n-2d+2} \dots c_n) : f(\mathbf{c}) = x\}$$

All the strings in C_x have length $2d-1$ and since these code suffixes have the same prefix, their minimum distance is at least d. Now, Plotkin bound is applicable on this subset of code words.

$$|C_x| \leq \frac{2d}{2d-(2d-1)} = 2d$$
$$\implies A_2(n,d) \leq 2^{n-2d+1} \cdot 2d = d \cdot 2^{n-2d+2}$$

The second equation comes from considering all possible values of x.

$$R = \lim_{n\to\infty} \frac{\log A_2(n,d)}{n} \leq \lim_{n\to\infty} \frac{\log d + n - 2d + 2}{n} = 1 - 2\delta$$

■

3 Johnson Bound

Applicable for constant weight codes: $A(n,d,w)$
Arrange the m code words along the rows of a $m \times n$ matrix.

$$\begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \vdots \\ X_1 & X_2 & .... & X_{n-1} & X_n \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \tag{6}$$

Denote by x_i the number of 1's in the i^{th} column.
Thus,

$$\sum_{i=1}^{n} x_i = mw$$

$$\begin{aligned} \binom{m}{2} d &\leq \sum_{i=1}^{n} x_i(m - x_i) \\ &= m^2 w - \sum_{i=1}^{n} x_i^2 \leq m^2 w - \frac{m^2 w^2}{n} \end{aligned}$$

$$\implies \frac{m(m-1)d}{2} \leq \frac{m^2 w(n-w)}{n}$$
$$\implies m \leq \frac{2nd}{(n-2w)^2 - n(n-2d)} \qquad \text{for } w < \frac{n - \sqrt{n^2 - 2nd}}{2}$$

4 Elias Bassalygo bound

Lemma 4 *For any set $C \subseteq \{0,1\}^n$ and for $x \in \{0,1\}^n$, define*

$$C + x = \{c + x : c \in C\}$$

Then,

$$\sum_{x \in \{0,1\}^n} |(C+x) \cap A| = |C| \cdot |A|$$

Proof

$$\begin{aligned}\sum_{x \in \{0,1\}^n} |(C+x) \cap A| &= \sum_{x \in \{0,1\}^n} \sum_{c \in C} \sum_{a \in A} \mathbb{1}_{\{x+c=a\}} \\ &= \sum_{c} \sum_{a} \sum_{x} \mathbb{1}_{\{x=a-c\}} \\ &= \sum_{c} \sum_{a} 1 = |C| \cdot |A|\end{aligned}$$

The second equality follows from the fact that for every pair (a, c), there is only one vector x such that $x = a - c$. ■

Corollary 5 *For $C, A, \exists x : |(C+x) \cap A| \geq \frac{|C| \cdot |A|}{2^n}$, where C is an (n, k) with minimum distance d, A is the set of all n-length vectors with Hamming weight w.*

The Elias Bassalygo bound uses the Johnson bound and the above lemma to develop a general bound for any code.

Theorem 6 *For a code with asymptotic rate R and asymptotic normalized distance d*

$$R \leq 1 - H\left(\frac{1 - \sqrt{1 - 2\delta}}{2}\right)$$

Proof With C and A as defined in the corollary to the lemma, $(C+x) \cap A$ is a constant weight code with minimum distance d. The size of this code $m = |(C+x) \cap A|$ is then upper and lower bounded as

$$\frac{|C|\binom{n}{w}}{2^n} \leq m \leq \frac{2nd}{(n-2w)^2 - n(n-2d)} \qquad \text{where } w \leq \frac{n - \sqrt{n^2 - 2nd}}{2}$$

$$|C| \leq \frac{2^n \cdot 2nd}{\binom{n}{w}\big((n-2w)^2 - n(n-2d)\big)}$$

Next, we choose $w = w_1$ such that $(n-2w)^2 - n(n-2d) = 1$. Then,

$$|C| \leq \frac{2^n \cdot 2nd}{\binom{n}{\theta n}} \qquad \text{where } \theta = \frac{w_1}{n} = \frac{n - \sqrt{n^2 - 2nd + 1}}{2n}$$

$$R = \lim_{n \to \infty} \frac{log|C|}{n} \leq \lim_{n \to \infty} \frac{n - log(2nd) - log\binom{n}{\theta n}}{n} = 1 - H(\theta)$$

Asymptotically, $\theta = \frac{1-\sqrt{1-2\delta}}{2}$, giving

$$R \leq 1 - H\left(\frac{1-\sqrt{1-2\delta}}{2}\right)$$

■